



Acid corrosion process of Fe-based bulk metallic glass

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ABSTRACT

For the bulk metallic glass ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) it is demonstrated that C plays a very important role in the acid corrosion process. Immersion in either 0.5 M H_2SO_4 or 1 M HCl yields constant-rate corrosion with the formation of a loose permeable C-rich layer. The underlying region of the corroding glass exhibits characteristic morphological features on the micro- and nanometre scale. The nanoscale features are explained by selective dissolution where C is the major slow component. The microscale features are attributed to a pitting-like process. It is demonstrated that *in situ* AFM is a powerful tool to visualize the incipient “micro-pit” evolution.

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1. Introduction

Fe-based metallic glasses are mostly known for their very good soft magnetic performance, extremely high corrosion resistance and high mechanical strength [1–3]. Despite these very good properties, their use was limited in the past owing to their small sample thickness that could be achieved by rapid solidification. During the last decade, progress in the design of alloy compositions led to the development of a new generation of Fe-based bulk metallic glasses (BMGs) with superior glass formation ability (GFA) and very good mechanical properties [4]. This makes them at present interesting as potential structural materials. However, another important aspect for the applicability of any material is its corrosion behaviour. Therefore, one of the most prominent Fe-based BMG with remarkably high critical diameter, i.e. 12 mm, and fracture strength of ~ 3 GPa [4] was selected for detailed corrosion investigations.

Numerous studies regarding the corrosion behaviour of Fe-based glasses have been carried out since the pioneering work of Naka et al. [5] demonstrating the superior corrosion resistance of several amorphous alloys compared to stainless steels. Most of them focused on developing highly corrosion-resistant alloys and indeed, extraordinary resistance was found for several alloys containing large amounts of Cr and Mo, e.g. Fe–25Cr–15Mo–18C with low GFA which passivates spontaneously in 12 M HCl at 60 °C [2]. Recently, research was mainly focused on characterizing the corrosion behaviour of alloys which were thermally-sprayed as coatings [6–8], on studying the influence of devitrification [9,10] and also on understanding the effect of compositional variations

on the pitting process [11]. Most of these studies focused on passivation and local corrosion processes. However, not so much attention was paid so far to active dissolution processes which are of particular importance for new Fe-based bulk metallic glasses with relatively low concentration of passivating elements, e.g. Cr. To better understand those processes in more detail phenomenological studies are needed. As well, not much attention was dedicated in previous studies to the role of metalloids in the corrosion process of Fe-based glasses although their concentration is quite significant, i.e. usually in the range of 20–30 at.%. It will be shown in this study that particularly C plays an important role.

For classical polycrystalline alloys, the morphology of corroding surfaces is governed by structural irregularities and chemical heterogeneities such as crystal orientation, dislocations, grain boundaries, secondary phases, etc. These determine locally varying corrosion rates. Surface sites which corrode at a higher rate will appear as valleys, while those corroding at a lower rate will appear as hills. Contrary to crystalline alloys, amorphous alloys ideally miss those structural and chemical particularities and therefore, they are considered as chemically and structurally homogeneous materials. The questions arise then: how is an active corrosion process on the surface of such a homogeneous amorphous alloy initiated and how does the morphology of such a corroding surface evolve? In order to analyze this, morphological studies on the micro- and nano-scale are required using *in situ* and *ex situ* observation techniques. Only few such studies have been reported for metallic glasses so far. For example, Green et al. [12] studied *in situ* with the help of optical microscopy the initial stages of pitting on $\text{Zr}_{50}\text{Cu}_{40}\text{Al}_{10}$ surfaces. In another study, Wang et al. [13] used atomic force microscopy (AFM) to characterize and compare the morphology of amorphous and crystalline $\text{Ni}_{50}\text{Nb}_{50}$ surfaces in

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order to understand the breakdown of passive films. In the case of Fe-based glasses, observations of morphology evolution of corroded surfaces have been limited mostly to *ex situ* SEM investigations of alloy surfaces mainly subjected to pitting or crevice corrosion [14–20]. Ha et al. investigated at the nano-scale the corrosion process of a TEM foil of the bulk metallic glass $\text{Fe}_{48}\text{Cr}_{15}\text{Mo}_{14}\text{C}_{15}\text{B}_6\text{Y}_2$ in 6 M HCl and observed preferential dissolution of chemical heterogeneities, i.e. nano-sized Y-Mo-rich islands [9].

In two of our previous papers we presented preliminary investigations on free corrosion, passivity and pitting of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}$) $_{98.5}\text{Y}_{1.5}$ alloy [21,22]. The passivation ability was found to be high in neutral and alkaline solutions, but rather low in acidic ones. The alloy exhibits high pitting resistance in neutral Cl^- -containing electrolytes. Nano-sized Y-oxide rich crystalline particles were identified and localized corrosion was preferentially initiated in the neighbouring amorphous matrix; however, this alloy exhibits a high resistance to further propagation of those localized dissolution events. The present work aims at analyzing in more detail the corrosion processes of this alloy in acidic electrolytes with special focus on the surface morphology evolution. It will be demonstrated that active dissolution reactions yield characteristic morphological features of the glassy alloy surface on the micro- and nanometre scale. For the first time, it will be shown that *in situ* AFM is a powerful method to study those corrosion phenomena for metallic glasses. It is detected that the features on the micro-scale are formed by a pitting-like process. To verify whether this process is influenced by Cl^- ions, investigations were conducted in solutions with and without Cl^- anions.

2. Experimental

2.1. Sample preparation

Ingots of nominal composition ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}$) $_{98.5}\text{Y}_{1.5}$ (at.%) were prepared by induction melting of pure Fe, Co, Cr, Mo, Mn, C and pre-alloys $\text{Fe}_{64}\text{B}_{54}$ and $\text{Fe}_{34}\text{Y}_{66}$ under purified argon atmosphere. From those ingots, rods with 3 mm diameter and 50 mm length were produced by injection casting into a water cooled Cu mould under argon atmosphere. Slices of 1 mm thickness were cut and used for the investigations described below (except for electrochemical polarization). After the preparation of the sample, its bulk glassy state was confirmed by X-ray diffraction (XRD), high resolution scanning electron microscopy (HRSEM), high resolution transmission electron microscopy (HRTEM) and differential scanning calorimetry (DSC). By HRSEM and HRTEM randomly occurring nano-sized Y-oxide particles embedded in the bulk glassy matrix have been observed. These crystalline defects and their effect on local corrosion processes have been reported in two previous papers [21,22].

2.2. Weight-loss measurements

The samples for weight-loss measurements were mechanically ground using emery SiC paper from grit 400 down to grit 4000 and then polished using 1 μm diamond suspension. Subsequently, the samples were ultrasonically cleaned in ethanol for 5 min and rinsed with distilled water. After their initial weight and surface area were measured, they were immersed in 0.5 M H_2SO_4 and in 1 M HCl. The samples were periodically taken out of the test solution (see Fig. 1 for the exact durations), rinsed with distilled water, weighted, microscopically investigated and re-immersed. The corrosion products were not removed at any of the intermediary stages. They were removed only at the end, after the last weight measurement, for microscopic investigations of the underlying al-

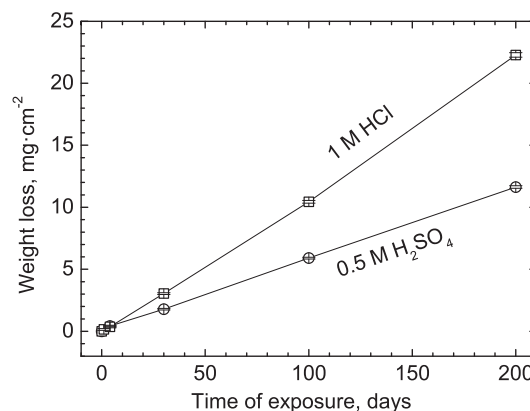


Fig. 1. Weight-loss measurement of bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}$) $_{98.5}\text{Y}_{1.5}$ alloy samples exposed to 0.5 M H_2SO_4 and 1 M HCl.

loy surface. For the weight measurements a Mettler Toledo AX205 analytical balance with the smallest increment of 0.01 mg was employed. Inductively coupled plasma mass spectrometry (ICP-MS) was employed for determining the amount of Fe, Cr, Co, Mo, Mn, B and Y released in the 0.5 M H_2SO_4 solution after 24 h. Five independent trials were conducted under similar conditions. The results presented here were determined by statistical analysis over the five experiments.

2.3. Electrochemical studies

For electrochemical studies a rod sample with 15 mm length was electrically connected to an isolated wire and embedded in epoxy resin so that only the cross-section remained free. Before each experiment, the sample surface was ground with emery SiC paper from grit 400 down to grit 4000 and then polished using 1 μm diamond suspension. The measurements were performed in 0.5 M H_2SO_4 (pH 0.3) and in 1 M HCl (pH 0.1). A Solartron SI 1287 electrochemical interface was employed. This was connected to a cell with three-electrode arrangement with a Pt net as counter electrode and a saturated calomel electrode (SCE) ($E = 241\text{ mV}$ vs. SHE) as reference electrode. The electrolytes were purged with N_2 for 1 h before and during each measurement. After monitoring the open circuit potential (OCP) for 1 h, potentiodynamic and potentiostatic polarization measurements in the anodic regime were performed. The dynamic polarization was started from the cathodic regime (about -100 mV vs. OCP) and linearly swept in the anodic direction at a rate of 0.2 mV s^{-1} until oxygen evolution set in or the current density reached a limit of 100 mA cm^{-2} . In order to have a high reliability of the results, all electrochemical measurements were repeated at least twice.

2.4. Microscopic investigations

After selected studies under open circuit or anodic polarization conditions, the morphology of the corroded surfaces was analyzed in detail. As it will be shown, immersion of the bulk glassy alloy for long periods of time, i.e. 200 days, in H_2SO_4 or in HCl generates a thick layer of corrosion products which obscure the sample surface. In order to remove it, the samples were ultrasonically treated for 5 min in bi-distilled water. Light optical microscopy (LOM), high resolution scanning electron microscopy (HRSEM) and (AFM) were employed. HRSEM was performed with a Gemini Leo 1530 instrument. AFM was conducted with a Park XE-100 instrument in non-contact mode.

2.5. X-ray photoelectron spectroscopy

Investigations of surface chemistry, e.g. elemental composition and oxidation state of elements, of corroded samples was carried out with X-ray photoelectron spectroscopy (XPS). This was performed with a PHI 5600 CI spectrometer (Physical Electronics). The hemispherical analyzer was operated at a pass energy of 25 eV. Monochromatized Al-K α radiation (1486.6 eV, 350 W) was used for excitation. To minimize the influence of surface charging, a low energy electron flood gun (approx. 1 μ A) was used. For comparison all the spectra were additionally corrected in binding energy with respect to the main C1s peak at 284.8 eV (C–C, C–H). Estimation of atomic concentration was done using standard single element sensitivity factors. Possible error by this simple calculation procedure and also due to inhomogeneities in the surface area should be considered. In order to perform analyses at various depths, the surface was sputtered on a 2×2 mm² area with argon ions, Ar⁺, with intensity of 2.5 μ A and acceleration voltage of 3.5 keV which gives a rate of material removal of about 2 nm min^{−1}.

2.6. In situ AFM observations

In order to study the initial stages of dissolution, *in situ* AFM investigations were performed. On top of an embedded sample a Teflon ring was placed; this acted as the wall of the cell. A three electrode setup was used in conjunction with a PS 6 (Sensortechnik Meinsberg GmbH) potentiostat. A Luggin capillary made of a Teflon tube was used to connect the reference electrode (SCE) to the cell. As counter electrode a Pt wire was used. During the measurements, the cantilever was fully immersed in the electrolyte, 0.5 M H₂SO₄. In order to be able to observe the very small initial topography changes, a very flat initial surface was required. For this, the sample was ground with emery SiC paper from grit 400 down to grit 4000 and then polished using 1 μ m diamond, 0.2 μ m and 0.04 μ m SiO₂ suspensions. Finally, the sample was cleaned with ethanol. The study was repeated to ensure reliability of the observations.

3. Results

3.1. Corrosion under open circuit conditions

3.1.1. Weight-loss analysis

Firstly, two disc samples of the bulk glassy (Fe_{44.3}Cr₅Co₅Mo_{12.8}Mn_{11.2}C_{15.8}B_{5.9})_{98.5}Y_{1.5} alloy were used for long-term weight-loss measurements in 0.5 M H₂SO₄ and 1 M HCl solutions. The samples have been periodically removed from the electrolyte for weight measurements and for surface analysis by SEM and AFM. The weight-loss values were calculated with respect to the initial surface area of the samples. As Fig. 1 shows, in both electrolytes the weight loss increases linearly with time indicating that the corrosion rate is constant in time. It can also be seen that the corrosion rate of this alloy is in 1 M HCl about twice as high as in

0.5 M H₂SO₄. Thus, after 200 days the weight-loss is in 1 M HCl 22.28 mg cm^{−2}, while in 0.5 M H₂SO₄ it is 11.63 mg cm^{−2}. As explained in Ref. [21] the corrosion rate is controlled by reduction of dissolved oxygen. The equilibrium potential of this reaction is pH dependent [23]. Therefore, the difference in weight-loss in 1 M HCl (pH 0.1) and 0.5 M H₂SO₄ (pH 0.3) is attributed to the different pH values of the solutions. It may be argued that the periodic removal of corroded samples from the test solution affects the corrosion process and therefore the corrosion rate. In order to verify this aspect, one sample was kept in 0.5 M H₂SO₄ for 200 days without interruption. The weight-loss of this sample (10.97 mg cm^{−2}) is close to the one when the process is interrupted (11.63 mg cm^{−2}). This proves that the periodic interruption has a minimal effect on the corrosion process.

Solution analysis by ICP-MS after 1 day of immersion in 0.5 M H₂SO₄ revealed that the bulk glassy alloy releases significant amounts of Fe, Cr, Co, Mo, Mn, B and Y. C cannot be analyzed by this method. Table 1 shows the amount released in solution as well as the calculated relative concentration of each element (among the elements considered above). The measured concentration values of the analyzed elements, except Mo and Y, match ($\pm 11\%$) the nominal ones. Less Mo seems to be released into solution than expected assuming equal dissolution rate for all alloy elements. However, as it will be demonstrated later, part of Mo transfers into the corrosion products layer. On the contrary, Y appears to be released at a higher rate than expected meaning it is preferentially dissolved. This does not come as a surprise as Y has a very large negative equilibrium potential at this pH value, i.e. ~ -2.5 V (pH 0.3), much larger than the other elements have.

3.1.2. Corroded surface topography

Microscopic SEM and AFM investigations of samples immersed in the two acid solutions revealed nearly identical surface morphological features. This indicates that the chloride ions do not play a critical role in the localization of the corrosion process. Therefore, only images of the surface of the alloy sample exposed to 0.5 M H₂SO₄ will be shown and discussed further. A series of representative SEM images is shown in Fig. 2. As it can be seen in Fig. 2(a), the as-polished surface is very smooth except for some scratches that were generated during the polishing process. After one day of exposure in 0.5 M H₂SO₄ (Fig. 2(b)), numerous nano-sized pores are observed. Their mean diameter is 12 nm. After 100 days, as shown in Fig. 2(c), round-shaped micrometer-sized pits are visible on the surface of the glassy alloy. Their maximum diameter is around 10 μ m, while their minimum diameter could not be determined precisely because of the presence of corrosion products. These pits are clearly distinguishable from the nano-sized pores visible in Fig. 2(b). After this exposure time (100 days), a change in the colour of the surface from shiny metallic to brown was observed. This indicates the presence of a corrosion product layer. This layer is (at this stage) electron-transparent when the classical SE detector and a high accelerating voltage, e.g. 20 kV, are used. This allowed visualization of the underlying alloy surface morphology. However, with an in-lens detector and a low accelerating

Table 1
Normalized amount of each element of bulk glassy alloy (Fe_{44.3}Cr₅Co₅Mo_{12.8}Mn_{11.2}C_{15.8}B_{5.9})_{98.5}Y_{1.5} (except for C) released in a period of 24 h in a 0.5 M H₂SO₄ solution at room temperature. Additionally, their relative atomic concentration values in solution along with the nominal ones (in alloy) are shown.

	Fe	Cr	Co	Mo	Mn	B	Y
Amount released in solution (μ g cm ^{−2})	24.0 ± 4.7	2.5 ± 0.6	2.4 ± 0.3	8.7 ± 1.1	5.2 ± 0.6	0.6 ± 0.1	2.0 ± 0.4
Relative concentration in solution (at.%)	55.13 ± 10.7	6.13 ± 1.5	5.21 ± 0.6	11.62 ± 1.5	12.07 ± 1.4	6.92 ± 0.8	2.93 ± 0.5
Nominal concentration (without C) (at.%)	51.7	5.8	5.8	14.9	13.1	6.9	1.8

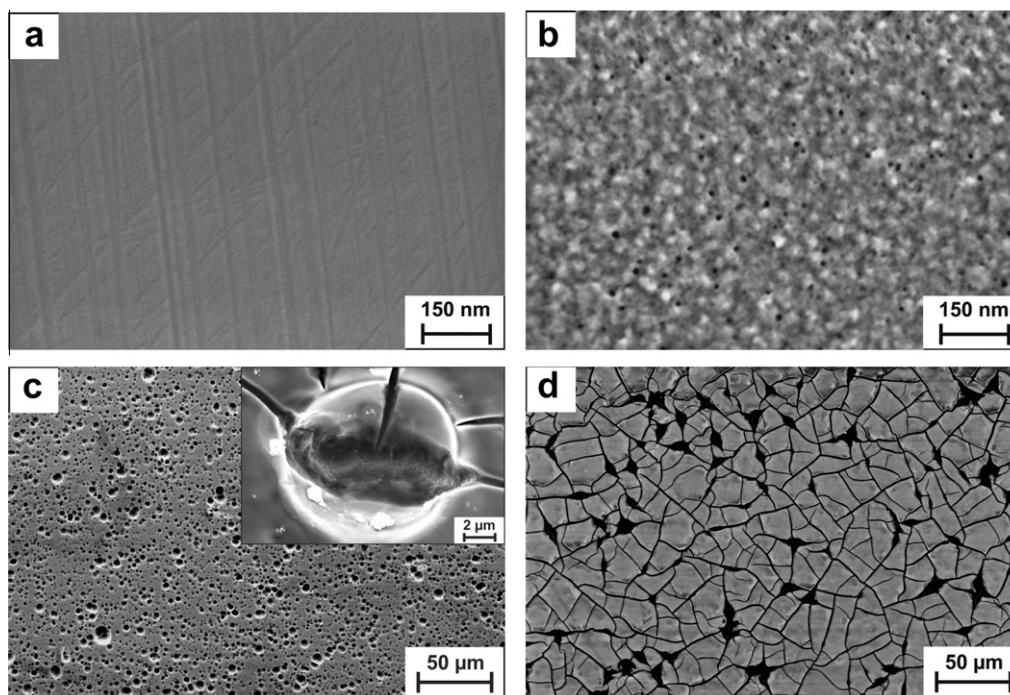


Fig. 2. SEM images of surfaces of bulk glassy $(\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9})_{98.5}\text{Y}_{1.5}$ alloy samples: (a) as-polished; after immersion in 0.5 M H_2SO_4 , (b) for 1 day, (c) for 100 days and (d) for 200 days. Inset in (c): close-up of a pit. Images (a), (c) and (d) taken with classical SE detector; images (b) and inset in (c) taken with in-lens detector.

voltage of the primary electrons, e.g. 5 kV, the layer is clearly visible as shown in the inset in Fig. 2(c). Further detailed morphological aspects seen in the inset of Fig. 2(c), especially those of the layer, are not taken into consideration: as explained later, removal of the samples from electrolyte and the high vacuum in the SEM device are altering the layer.

After 200 days of exposure, the alloy surface changed to black colour corresponding to an increase of the corrosion product layer thickness which made it light-and electron-opaque. A characteristic SEM image of the alloy surface at this stage is shown in Fig. 2(d). The corrosion product layer is not continuous, but contains many cracks which formed as a consequence of the drying process of the originally hydrated layer after removal from the electrolyte and insertion in the SEM. Measurements with SEM revealed a mean thickness of the corrosion product layer of approximately 2 and 4 μm after 200 days in 0.5 M H_2SO_4 and 1 M HCl, respectively. The layer thickness increases with time and since the corrosion rate is constant, as seen from the weight-loss curve in Fig. 1, it can be concluded that diffusion of any species, e.g. diffusion of ions, through this layer is not the rate-controlling step of the total corrosion process.

More detailed SEM investigations of the sample exposed to 0.5 M H_2SO_4 have been performed after removal of the corrosion product layer in order to characterize the underlying alloy surface morphology. As Fig. 3(a) shows, after 200 days of exposure the surface of the sample is completely covered with round-shaped pits with size values from 0.2 to 12 μm .

This wide size distribution suggests that while early formed, larger pits grow further, new smaller pits form, implying a progressive nucleation process. New pits appear randomly—there are apparently no preferred surface sites for the nucleation of new pits. At higher magnification it is obvious that, as visible in Fig. 3(b), the surface within a micro-pit exhibits a nano-porous structure. This substructure occurs similarly on the walls of all micrometre-sized pits. The mean pore size of the substructure is 15 nm, which is similar to the mean pore size determined after one day of exposure, i.e. 12 nm (see also Fig. 2(b)). It is therefore concluded that the same process which initially generated this nano-porous structure continues to take place at the later stages when the micrometre-sized pits start to form and grow. These observations suggest that although the two processes, i.e. the one causing the nano-porous structure and the one causing the micrometre-sized pits,

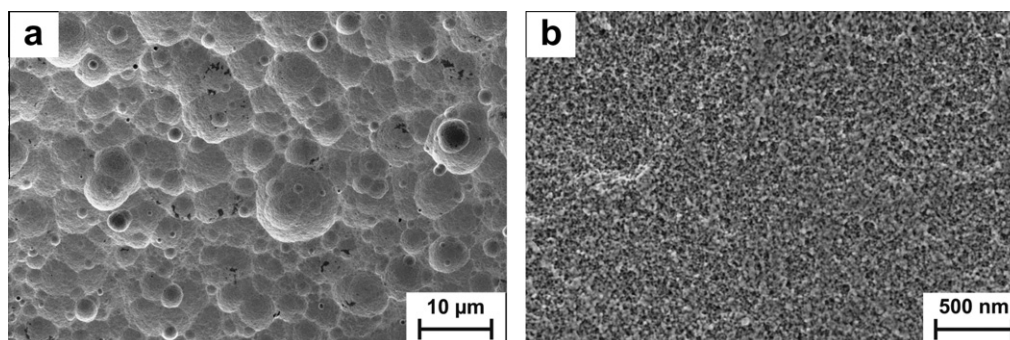


Fig. 3. SEM images (secondary electron mode) of surfaces of a bulk glassy $(\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9})_{98.5}\text{Y}_{1.5}$ alloy sample after 200 days exposure to 0.5 M H_2SO_4 . The corrosion layer was removed by ultrasonic treatment in bi-distilled water. (a) Low magnification—"micro-pits", and (b) higher magnification—"nano-porosity".

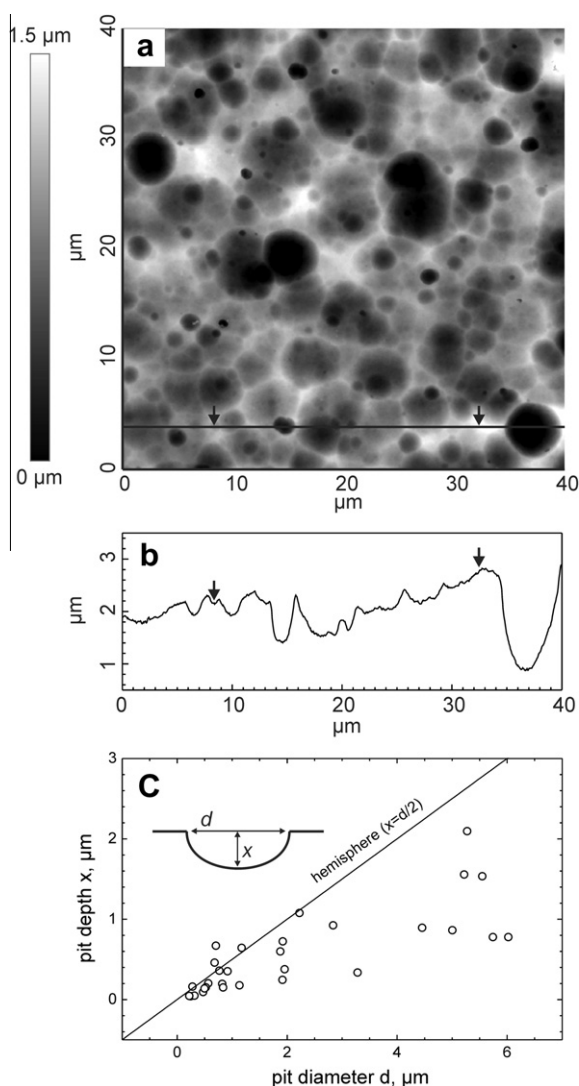


Fig. 4. (a) Non-contact AFM image of the surface of a bulk glassy alloy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$ sample after 200 days of exposure to 0.5 M H_2SO_4 , (b) line scan corresponding to the line indicated in image (a), and (c) depth versus width of 30 pits selected from image (a).

take place simultaneously, they proceed independently one from each other.

Further investigations of the corroded surface morphology were conducted by means of AFM. Fig. 4(a) shows an AFM image of a glassy alloy surface which was exposed for 200 days to 0.5 M H_2SO_4 and subsequently cleaned from corrosion products. As already observed by SEM (see Fig. 3(a)), round-shaped micro-pits with various sizes are present. The diameter of the pits measured by AFM varies between 200 nm and 6 μm . As compared to SEM, AFM provides much more detailed information on the surface topography. Pit dimensions are determined more accurately, especially in the vertical direction [24]. Indeed, as it can be seen in the line profile in Fig. 4(b), the shape of pits in the vertical direction is very well defined. In order to characterize the general shape of the pits, the diameter and depth of 30 pits representing the whole size interval were determined from such line profiles. Fig. 4(c) shows a plot in which the diameter and the depth of those pits are represented. In order to compare their shape to that of a perfect hemisphere, a corresponding theoretical line is added. It is evident that the shape of the majority of pits is semi-ellipsoidal, the short axis being perpendicular to the surface plane, i.e. in depth. This

indicates that the corrosion front propagates on the glassy alloy surface more laterally than in depth resulting in shallow pits. Unfortunately, the nano-porous substructure visible in the HRSEM image in Fig. 3(b) could not be satisfactorily imaged with AFM due to tip-sample convolution.

3.1.3. XPS analyses

For the determination of the elemental composition and the oxidation state of each element in the corrosion product layer of the bulk glassy samples corroded in 0.5 M H_2SO_4 and in 1 M HCl, XPS analyses were performed on the as-corroded surface as well as after 2 min of sputtering. In all the resulting spectra, peaks corresponding to C, O and Mo were detected (the measured spectra are not shown here). Quantification of high resolution detail spectra in the region of these three elements resulted in the concentration values indicated in Table 2. The corrosion products layer on samples exposed to the two solutions comprises mainly C and O. In the as-corroded state (no sputtering) there are two peaks corresponding to C1s: a major one at 284.8 eV which is associated with the simple C–C bonding and a minor one at 288.8 eV assigned to the C bonding in carboxyls (assignment of these peaks and of the others following, unless otherwise indicated, are done based on Ref. [25]). The O 1s spectrum shows a single peak at 532.3 eV which also indicates the presence of carboxylic groups [26]. A second series of spectra was determined after sputtering the surface for 2 min which corresponds to a depth of approximately 4 nm. The most important difference is the O concentration, i.e. it is approximately 4 times lower after sputtering (see Table 2). The small quantity of Mo appears to be in an oxidized state corresponding to MoO_3 and MoO_2 . It is concluded that the corrosion product layer is mostly comprised of C. Most of the C atoms form C–C bonds. However, additionally the presence of a significant amount of O-containing groups (C–O bonds) is noted.

Next, to analyze the alloy surface, the corrosion products were removed from the sample surface by ultrasonic cleaning in bi-distilled H_2O . A first set of XPS spectra was determined immediately to prevent any changes in the surface composition. Subsequently, two other sets of spectra were determined after sputtering with Ar ions for 2 and 7 min respectively. Additionally, as a reference for comparison, to reflect the bulk of the alloy, a set of spectra was acquired after 30 min of sputtering. Further, quantification of all these spectra was carried out. The resulting concentration values of each element are plotted versus sputtering time (reflecting depth from surface) in Fig. 5. The non-sputtered surface of both samples after removal of the corrosion product layer has a composition similar to the one of the layer except for very small amounts of additional metallic elements, i.e. Fe, Co and Mn. This might be due to the presence of corrosion products remainings. The underlying non-sputtered alloy surface has however higher concentration of O and lower concentration of C compared to the layer. This might be explained by the presence of an underlying oxide film as described below. After just 2 min of sputtering, i.e. equivalent to approximately 4 nm depth, major changes are evident. The

Table 2

Surface elemental composition of corrosion products layer on bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$ alloy samples after 200 days in 0.5 M H_2SO_4 and 1 M HCl—as determined by quantification of XPS spectra.

Solution	Sample condition	Concentration (at.%)		
		C	O	Mo
0.5 M H_2SO_4	No sputtering	71	29	0.4
	Sputtered 2 min	92	7	1.3
1 M HCl	No sputtering	72	27	0.4
	Sputtered 2 min	92	7	0.9

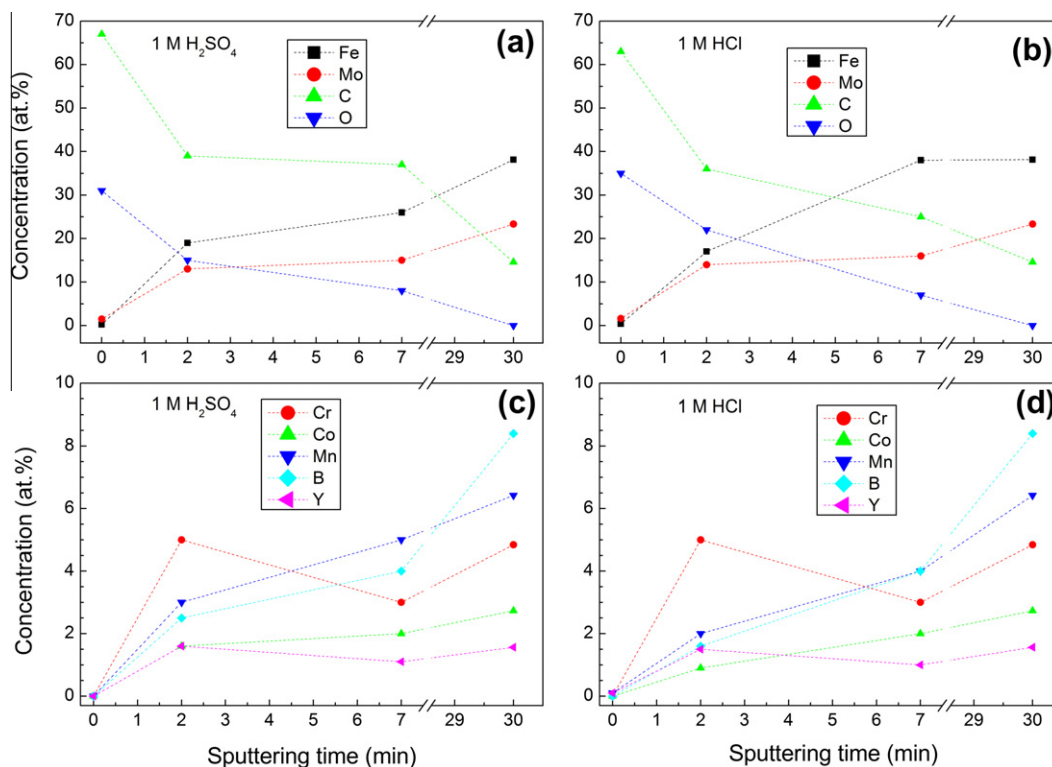


Fig. 5. Concentration-depth profiles from XPS analyses on the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy samples exposed for 200 days to 0.5 M H_2SO_4 (a, c) and 1 M HCl (b, d) respectively (after removal of the corrosion product layer).

C and O concentration values decrease while that of metallic elements increase drastically. Additionally, significant changes occur in the oxidation state of the elements as shown by the shift in their binding energy. In the case of C, the carboxyl-specific peak is no longer distinguishable and correspondingly, the O 1s peak shifts from the carboxyl-specific position to values (531.0 eV) that are usual for metallic oxides, e.g. Cr_2O_3 531.0 eV. While Mo, Fe, and Co show after 2 min no visible peak corresponding to an oxidized state, Cr, Mn and B show oxide-specific peaks with high intensity, e.g. $\text{Cr}(\text{OH})_3$, Mn_2O_3 , B_2O_3 . These observations indicate the presence of a thin nano-sized film of mixed oxides of Cr, Mn and B. Correspondingly, Cr appears to be enriched at this depth, i.e. 2 min sputtering (see Fig. 5). Similar to Cr, Y appears to be enriched at the depth corresponding to 2 min of sputtering. However, its peak position could not be associated with any compound previously investigated by XPS. With increasing depth from surface, the O concentration decreases and after a very long sputtering period, i.e. 30 min, it is no longer detectable indicating that at this depth the alloy is not affected by the corrosion process. At this level, the measured concentrations of the alloy elements are close to the nominal ones and their peak position correspond to carbide- and boride-like states. This observation correlates with the short range order of metal-metalloid glasses consisting of metalloid-centered clusters similar to the ones in carbides [27].

As Table 2 and Fig. 5 show, corrosion in the two solutions generates very similar composition of corrosion products and underlying alloy surface. This indicates that both solutions yield similar corrosion products on the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy. In conclusion, the XPS investigations of corroded samples of the bulk glassy alloy after 200 days exposure to 0.5 M H_2SO_4 and 1 M HCl respectively, revealed a complex surface chemistry. On the surface, a few micrometres thick layer of corrosion products mainly composed of C and O is present. The underlying alloy is covered by a thin (few nanometers) film of Cr, Mn and B oxides.

3.2. Corrosion under anodic polarization conditions

Analysis of anodic corrosion processes in direct relation with the surface morphology evolution of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy was performed by means of potentiodynamic and potentiostatic polarization measurements in conjunction with microscopic observations. Potentiodynamic polarization investigations were conducted in 0.5 M H_2SO_4 (pH 0.3) and 1 M HCl (pH 0.1) and typical curves are shown in Fig. 6. In 1 M HCl, the corrosion potential is more negative than in 0.5 M H_2SO_4 , i.e. $E_{\text{corr}}^{\text{HCl}} = -198$ mV compared to $E_{\text{corr}}^{\text{H}_2\text{SO}_4} = -37$ mV. Also, the corrosion current density is higher, i.e. $i_{\text{corr}}^{\text{HCl}} = 0.01$ mA cm^{-2} compared to $i_{\text{corr}}^{\text{H}_2\text{SO}_4} = 2.3 \times 10^{-4}$ mA cm^{-2} . This indicates that the bulk glassy alloy exhibits higher reactivity in 1 M HCl than in 0.5 M H_2SO_4 . However, during anodic polarization at potentials more positive than 200 mV, the alloy exhibits a quite similar response in both electrolytes. The overall anodic current density level is high, i.e. it reaches values of 2 and 8 mA cm^{-2} at 1000 mV in 0.5 M H_2SO_4 and 1 M HCl, respectively. Additionally, for the bulk glassy alloy a potentiostatic measurement was performed at 600 mV for 1 h in 0.5 M H_2SO_4 . The resulting potentiostatic curve is shown in the inset of Fig. 6. The average value of current density during this potentiostatic measurement, i.e. 1.17 mA cm^{-2} , corresponds very well to the value registered during the potentiodynamic polarization at the same potential (600 mV), i.e. 0.85 mA cm^{-2} . The current density value varies in a small range during the 1 h measurement period, i.e. 1.10–1.23 mA cm^{-2} . This variation can be regarded as insignificant, but an overall trend of increase with time can be claimed after 2400 s. This increase in current density is explained by the increase of the reactive surface area. As it will be seen later, this increase is the result of roughening of the exposed surface. Furthermore, it is noted that this increase in current is not uniform, but occurs step-wise. These abrupt variations indicate the occurrence of processes with very high rate which are typical for pitting initiation. As it will be shown

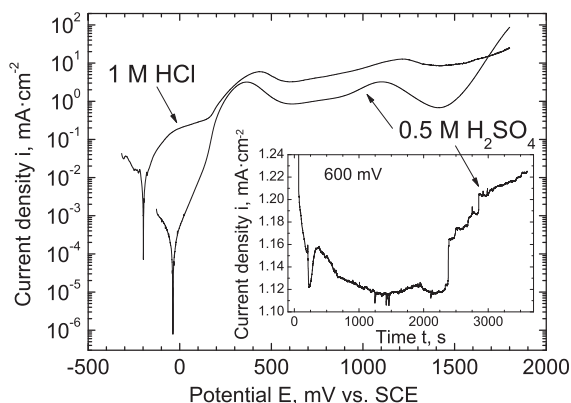


Fig. 6. Potentiodynamic polarization curves of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy in 0.5 M H_2SO_4 and in 1 M HCl. Inset: potentiostatic polarization curve of the same alloy at 600 mV vs. SCE in 0.5 M H_2SO_4 .

in the next paragraph, pits are indeed observed (see Fig. 7(c)). These abrupt variations are therefore believed to correspond to the formation of those pits.

Light optical microscopy (LOM) investigations of the alloy surface performed after the complete polarization in the 0.5 M H_2SO_4 electrolyte (see Fig. 7(a)) revealed a similar type of surface morphology as observed in the case of corrosion under open circuit conditions (Fig. 2(c)), i.e. after 100 days of exposure to 0.5 M H_2SO_4 , where uniformly distributed round-shaped pits were found to be characteristic. This is similarly the case after complete polarization in 1 M HCl (see Fig. 7(b)) and after 1 h potentiostatic polarization at 600 mV in 0.5 M H_2SO_4 (see Fig. 7(c)). There are fewer pits formed after potentiostatic polarization in 0.5 M H_2SO_4 than after complete potentiodynamic polarization in either of the two electrolytes. This is because less charge is consumed in this potentiostatic measurement. However, the principal pit morphology is similar as in all the other cases. These observations lead to the conclusion that the morphology of the actively corroded surface is not determined by the electrolyte or by the potential regime.

As mentioned earlier, the samples kept at OCP were slowly covered with a C-rich layer in both 0.5 M H_2SO_4 and 1 M HCl. On the contrary, no corrosion products layer could be seen by LOM or SEM on either of the polarized samples. As it will be seen later in the discussion section, this can be explained by the oxidation of the C atoms to form CO_2 .

The same ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy, but in the shape of a melt-spun ribbon was also potentiodynamically polarized and a very similar corrosion morphology was developed in this case also. Compared to the bulk glassy alloy sample which was polished before each experiment, the ribbon sample surface was initially used in the as-quenched state. The similarity of the surface morphology of these two samples after the corrosion investigations indicates that the initial surface state is also not a decisive factor for the corrosion mode.

In conclusion, the micro-scale surface morphology evolution of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy during acid corrosion is neither determined by the nature of the acid electrolyte, i.e. H_2SO_4 or HCl, nor by the initial surface state or the applied polarization regime. It is therefore concluded that the corrosion morphology is most likely determined by material-specific characteristics such as alloy composition and structure.

3.3. Initial corrosion stages

In order to analyze the initial stages of pit formation during metallic glass corrosion in acid solutions, *in situ* AFM investigations

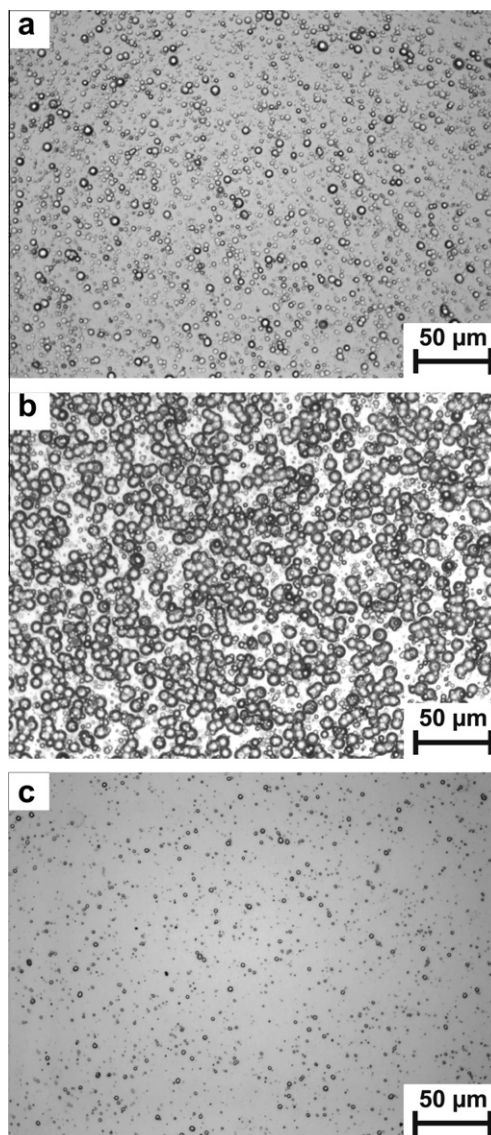


Fig. 7. LOM images of surface of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy sample after: (a) complete polarization cycle in 0.5 M H_2SO_4 , (b) complete polarization cycle in 1 M HCl, and (c) 1 h potentiostatic polarization in 0.5 M H_2SO_4 at 600 mV vs. SCE.

were performed in 0.5 M H_2SO_4 . For this, the glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) sample was firstly exposed for 1 h under open circuit conditions in the *in situ* AFM cell and then an anodic potential of 600 mV vs. SCE was applied. AFM images were taken before applying the anodic potential and then periodically, every 10 min during polarization up to a total polarization time of 60 min. During the AFM imaging, the anodic polarization was interrupted. Selected AFM images are shown in Fig. 8. It can be seen that already before applying the anodic potential, i.e. after 1 h under open circuit conditions, single small pits with diameter of about 200 nm and mean depth of only about 5 nm are visible (Fig. 8(a)). After 10 min of polarization at 600 mV vs. SCE, the same pit (it appears at a different site in the image because the sample drifted with time) has grown in diameter and in depth. It can be seen in Fig. 8(b) that at this time the pit is 280 nm wide and 14 nm deep. An additional small pit appeared at the rim of this one. After 60 min (Fig. 8(c)) the same pit grew further reaching 800 nm in diameter and 143 nm in depth. Furthermore, new pits appear progressively at other surface sites. It is noticed that the ini-

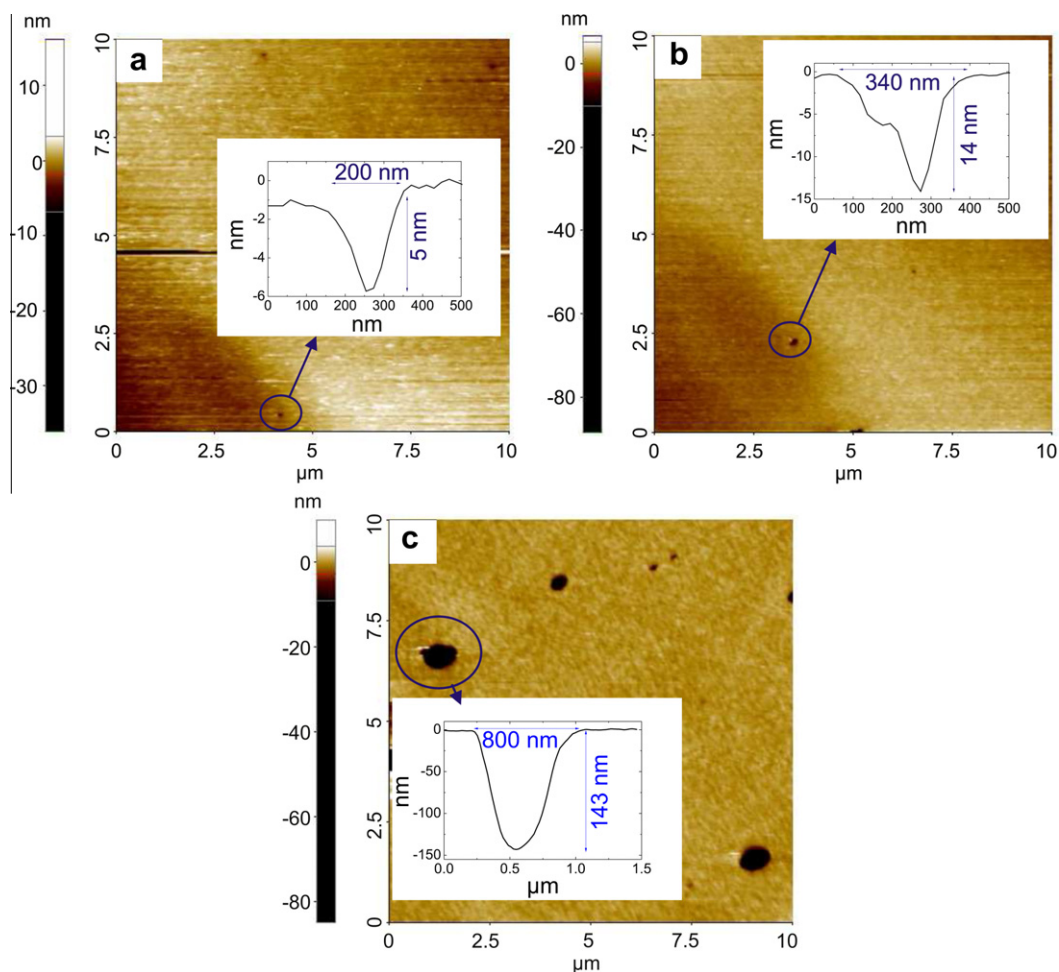


Fig. 8. *In situ* AFM images of the surface of a bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy sample surface in 0.5 M H_2SO_4 (a) at the beginning of the polarization measurement (after 60 min at OCP), (b) after 10 min at 600 mV vs. SCE, and (c) after 60 min at 600 mV vs. SCE.

tially very flat pits, e.g. depth:diameter = 5:200 nm (Fig. 8(a)) grow faster in depth than in diameter resulting in nearly hemispherical shapes, e.g. 143:800 nm (Fig. 8(c)). Both AFM modes, *in situ* and *ex situ*, show that the bigger pits, i.e. $>1\ \mu\text{m}$, have a shape tending to hemispherical. However, in the case of the smaller pits, while *ex situ* AFM reveals a similar shape, *in situ* AFM reveals much flatter shapes. This could be caused by the partial filling of the newly created pit with corrosion products. It is demonstrated that *in situ* AFM has the capability of observing in detail the evolution of the corroding surface of the glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy sample at a fixed location without taking the sample out of the test electrolyte. *Ex situ* investigations with AFM or SEM would require much more effort and would have the additional disadvantage of taking the sample out of the test solution for every investigation.

4. Discussion

Based on the experimental observations, a description of the surface morphology evolution during active corrosion in acid solutions of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy can be given. A schematic illustration of the process is shown in Fig. 9. Upon immersion in the electrolyte, two processes take place as shown in the second cartoon for $t = 24\ \text{h}$. First, as shown by the SEM image in Fig. 2(b) changes in surface morphology are taking place leading to formation of nano-pores. Second, dissolution of Fe, Cr, Co, Mo, Mn, B and Y occurs as revealed by ICP-MS.

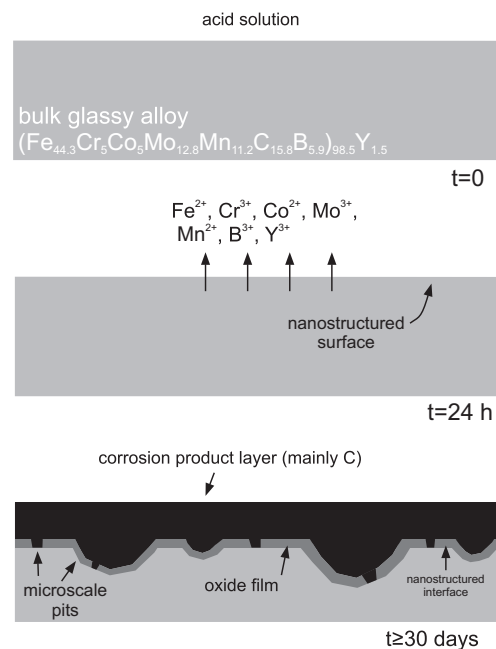


Fig. 9. Schematic illustration of the surface morphology evolution of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{1.5}$) alloy during corrosion in acid solutions at open circuit conditions.

There is no proof that C is also dissolving into the solution. XPS analyses on the alloy surface after 200 days in either 0.5 M H_2SO_4 or 1 M HCl prove that C tends to resist being released from the alloy and is therefore enriched at the surface. It is assumed that such a resistance is also shown at an earlier time, at the beginning of the corrosion process, e.g. 24 h after immersion. Therefore, it is believed that this selective dissolution process causes the surface nano-scale roughening by a mechanism similar to the one characteristic to 'higher-melting' crystalline alloys (implying a low diffusivity at room temperature) [28]. After 200 days of exposure, also Cr and Y seem to be slightly enriched at the alloy surface meaning they also, as C, act as slow components in the dissolution process. However this is not true for the initial stage (before the C layer is formed) – as ICP-MS shows, in the first 24 h Cr and Y are dissolved faster. With progressing time, C accumulates at the surface and forms a relatively loose uniform layer which also contains a significant amount of O species, i.e. 25–28 at.% at surface (7 at.% below surface), bound to C atoms. XPS revealed that the O is bound to part of the C as in carboxylic groups. This indicates that part of C from the alloy is oxidized to carboxyl C. However, it has to be kept in mind that removal of the sample from the electrolyte and the high vacuum in the XPS device might alter the corrosion products layer. Nevertheless, due to the presence of O-containing groups, the C layer is expectedly hydrophilic and therefore can be soaked [26]. The formation of cracks in the layer upon drying and further exposure to high vacuum inside the SEM chamber (see Fig. 2(d)) points out that it contains water or some other water-based solution. Additionally, a fully dense pure C layer would not be expected to have the ability to allow the transport of the reaction products or at least not at the high rates that were observed. Instead, a gel structure or a porous solid structure that allows direct access of the test electrolyte to the surface of the alloy seems more feasible. As SEM investigations revealed, this C-rich layer thickens with time. Since transport of species involved in the corrosion process through this layer is not the rate-limiting step of the overall corrosion process, it can thicken indefinitely without affecting the corrosion rate. Such a C rich layer was not observed after corrosion at higher potentials because the C atoms are expected to be oxidized to CO_2 [29,30].

In addition to the nanostructuring of the alloy surface and to the formation of the C layer, formation of round-shaped pits on the micro-scale was observed, i.e. after 1 h in 0.5 M H_2SO_4 (Fig. 9, third cartoon). Two mechanisms are proposed for the formation of those pits. The first mechanism is based on the classical mechanism of pitting corrosion of passivated metal surface [31]. As revealed by the XPS analysis, the surface of the alloy (underneath the corrosion product layer) is covered by a very thin film of Cr-, Mn- and B-oxides. This film could act as a passive film although not very protective since dissolution rates remain quite high. However, it is not clear why some surface sites are more favourable than others for "pitting" initiation since the investigated alloy is amorphous and therefore considered to be ideally homogeneous. As well, it is not clear, yet, why continued higher dissolution rate persists at those locations until the pit grows in the micrometer range. The second mechanism is presented in the next paragraph, but first an analogy of the investigated alloy with crystalline carbides is made.

Since C plays a significant role in the corrosion behaviour of this bulk glassy alloy, a discussion focusing on this is very pertinent. Payer and Staehle did a thorough study of the role of C in dissolution processes of crystalline carbides [29]. They observed that during dissolution of metal carbides, C can be reduced to hydrocarbons, oxidized to C oxides or remain on the surface as free C depending on the carbide structure and the potential at which dissolution occurs. Metal-metalloid (M-Me) type glasses resemble common crystalline carbides (and borides for that matter) with regard to the metal to metalloid concentration ratio, i.e. 20–25 at.%

Me in metallic glasses, and 20.7 at.% C in Cr_{23}C_6 and 25 at.% C in Fe_3C . As well, they have similar local structural units (local atomic arrangements on the short range scale), i.e. trigonal prismatic units with a metalloid atom at the centre [27,32,33]. Therefore, a qualitative comparison with regard to dissolution behaviour between the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{98.5}\text{Y}_{1.5}$) alloy and crystalline carbides is not trivial. In consequence, hydrocarbons, e.g. methane, CH_4 , are expected to form on this bulk glassy alloy under reducing conditions [29]. Under oxidizing conditions, or at higher potentials, the C atoms are expected to oxidize to C oxides such as CO and CO_2 [29,30]. At potentials which are not notably favoring one or the other product, both of them, i.e. CH_4 and CO_2 , can form and additionally free C is produced on the surface [29]. Therefore, it might be expected that at open circuit conditions in addition to the observed solid C rich layer, also gaseous products such as CH_4 and CO_2 are formed. This would in turn lead to formation of gas bubbles of those products. A new mechanism for the formation of the observed pits on the micrometre scale, different from the one proposed earlier in this discussion, can then be proposed. Assuming that those bubbles form at the interface in-between the C rich layer and the underlying glassy alloy, they will burst when reaching a critical pressure, thus breaking the overlying corrosion product layer. This way the underlying base alloy surface is directly exposed to the acid electrolyte which might lead to a more active reaction than that when the gel-like electrolyte has contact with the surface. This can lead to the formation of a pit. The pit growth will stop when sufficient corrosion products have been produced inside the pit and cover the pit walls. At higher potentials at which formation of pits also takes place, the C layer is not present. Instead an (hydro) oxide layer (of Fe and Cr at low anodic polarization and of Mn and Co at high polarization) with poor protective ability is covering the surface [22]. Since C-species are scarcely soluble in oxides [34] and therefore C atoms from the alloy are not expected to diffuse through the oxide film, they can be oxidized to CO_2 below the oxide film. This further sustains the idea that CO_2 bubbles formation mechanically break the oxide film, thus exposing locally the base alloy surface to the aggressive electrolyte. Rapid active dissolution occurs at the respective location causing the formation of a pit and a corresponding abrupt increase in the measured current density, e.g. see inset of Fig. 6. The role of the other constitutive elements of the bulk glassy alloy, i.e. other than C, depends on the polarization conditions: when no polarization is applied, i.e. free corrosion, those elements dissolve into solution (Cr, Mn and B are partially retained at the surface as oxides) while under anodic polarization they form various (hydro) oxides.

A very similar surface morphology evolution was observed by Shan et al. in the case of the bulk glassy SAM 1651 alloy ($\text{Fe}_{48}\text{Cr}_{15}\text{Mo}_{14}\text{C}_{15}\text{B}_6\text{Y}_2$) in a crevice corrosion measurement in 4 M NaCl at 100 °C at a potential of 150 mV vs. SCE [16]. In that case, the growth of larger pores was explained to be initiated at the location of the Y_2O_3 particles. Y_2O_3 particles are also present in the alloy studied in the present work [21,22], but their number is much lower than the number of observed pits. This leads to the conclusion that the described surface morphology evolution during acid corrosion is not directly related to the presence of those secondary phase particles, as was the case in the study by Shan et al. Moreover, it was observed that most of these particles located at the surface were removed during the surface polishing process [22].

5. Conclusions

The complex corrosion process of the bulk glassy ($\text{Fe}_{44.3}\text{Cr}_5\text{Co}_5\text{Mo}_{12.8}\text{Mn}_{11.2}\text{C}_{15.8}\text{B}_{5.9}\text{Y}_{98.5}\text{Y}_{1.5}$) alloy in acidic solutions yields characteristic surface morphological features on the micro- and

nanometre scale. Carbon from the alloy forms a loose permeable layer on top of the corroding surface while all the other constitutive elements are dissolved into solution. Additionally, on the surface of the corroding alloy (below the carbon rich layer) a film of mixed Cr, Mn and B oxides was detected by XPS. The features on the nano-scale can be explained by a selective dissolution process where C is the major slow component. The features on the micro-scale are thought of being caused by a pitting-like process where the thin oxide layer acts as a pseudo-passive film. The initial breakdown of this film might be caused by the sudden direct exposure of the alloy surface to the electrolyte subsequent to local rupturing of the C-rich layer by growing CO₂ bubbles. Under anodic conditions, a C rich layer is not observed. Rather, C is expected to be oxidized to CO₂. Instead, the surface is covered by a not very protective layer of (hydro) oxides. Formation of the micro-scale features can be caused by a process similar to the one without applied anodic polarization.

In conclusion, the bulk glassy (Fe_{44.3}Cr₅Co₅Mo_{12.8}Mn_{11.2}C_{15.8}B_{5.9})_{98.5}Y_{1.5} alloy exhibits a complex corrosion process in acidic environments which is explained mainly by the high concentration of C in this alloy and its particular composition and structure, i.e. the homogeneous distribution of C atoms in the amorphous SRO structure. On the example of this multi-component bulk glass-forming alloy the principle acid corrosion process has been fundamentally studied. It can be well predicted that the proposed model for the evolution of the surface morphology, and the related corrosion reactions comprising the special role of C, can be transferred to other Fe-based glass-forming systems.

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